Initial sum	0001	0010	0011	1010	1100	1011	1110
00-1	X		X				
11-0					X		X
1-10				X			X
-01-		X	X	X		X	

1-10 is superflous prime implicant and can be left.

$p \sim x'y + wxz' + w'x'z$
is the required minimal form of p

10/23
Ques. Determine the minimal form of the polynomial p, which is given in DN-form.

$$p = v'w'x'y'z' + v'w'x'yz' + v'w'xy'z' + v'w'xyz'$$
$$+v'wx'y'z + v'wx'yz' + v'wxy'z + v'wxyz'$$
$$+v'wxyz + vw'x'y'z' + vw'x'y'z + vw'xy'z$$
$$+vwx'yz' + vwxy'z' + vwxyz' + vwxyz$$

$v'w'x'y'z' =$ 00000 — 0 ones

$v'w'x'yz' =$ 00010 — 1 ones

$v'w'xy'z' =$ 00100 — 1 ones

$v'w'xyz' =$ 00110 — 2 ones

$v'wx'y'z =$ 01001 — 2 ones

$v'wx'yz' =$ 01010 — 2 ones

$v'wxy'z =$ 01101 — 3 ones

$v'wxyz' =$ 01110 — 3 ones

$v'wxyz =$ 01111 — 4 ones

$vw'x'y'z' =$ 10000 — 1 ones

$vw'x'y'z =$	10001	2 ones
$vw'xy'z =$	10101	3 ones
$vwx'yz' =$	11010	3 ones
$vwxy'z' =$	11100	3 ones
$vwxyz' =$	11110	4 ones
$vwxyz =$	11111	5 ones

00000	(1)
00010	(2)
00100	(3)
10000	(4)
00110	(5)
01001	(6)
01010	(7)
10001	(8)
01101	(9)
01110	(10)
10101	(11)
11010	(12)
11100	(13)
01111	(14)
11110	(15)
11111	(16)

000-0	(1)(2)	
00-00	(1)(3)	
-0000	(1)(4)	——— (A)
00-10	(2)(5)	
0-010	(2)(7)	
1000-	(4)(8)	——— (B)
001-0	(3)(5)	

0-110	(5)(10)	
10-01	(8)(11)	——— (C)
01-01	(6)(9)	——— (D)
01-10	(7)(10)	
-1010	(7)(12)	
011-1	(9)(14)	——— (E)
-1110	(10)(15)	
111-0	(13)(15)	——— (F)
0111-	(10)(14)	
11-10	(12)(15)	
-1111	(14)(16)	
1111-	(15)(16)	

00-0	(1)(3)(2)(5)
00-0	(1)(2)(3)(5)
0-10	(2)(5)(7)(10)
0-10	(2)(5)(7)(10)
-1-10	(7)(12)(10)(15)
-1-10	(7)(10)(12)(15)
-111-	(10)(15)(14)(16)
-111-	(10)(14)(15)(16)

Rewrite the above table by removing duplicates.

00-0	(1)(2)(3)(5)	——— (G)
0-10	(2)(5)(7)(10)	——— (H)
-1-10	(7)(10)(12)(15)	——— (I)
-111-	(10)(14)(15)(16)	——— (J)